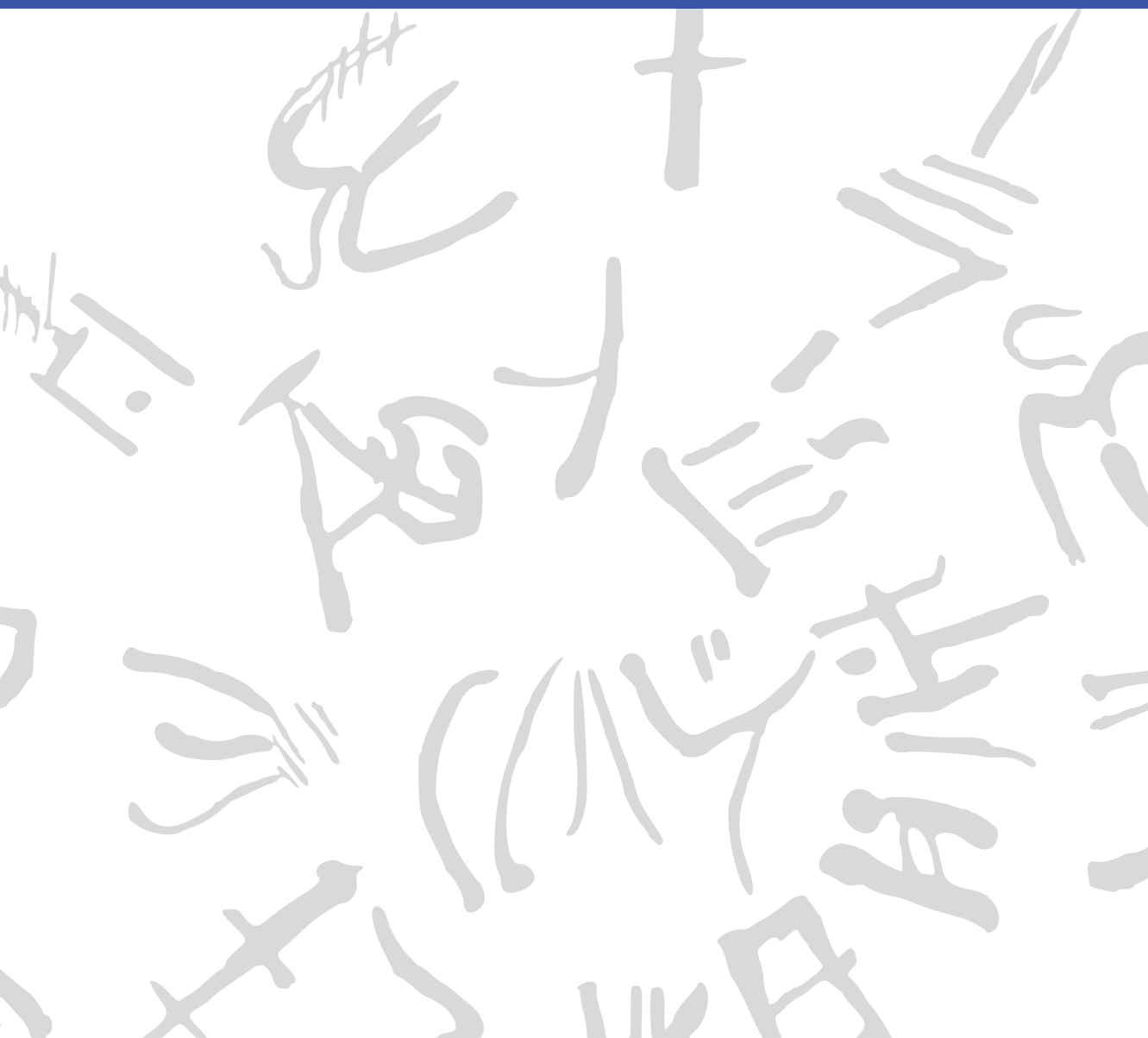


Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

Bachelor Thesis



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Preface

This Bachelor's thesis was prepared at the Department of Applied Mathematics and Computer Science at the Technical University of Denmark in fulfillment of the requirements for acquiring a Bachelor of Science in Engineering (BSc Eng) degree in General Engineering (Cyber Systems).

Kongens Lyngby, March 30, 2026

A handwritten signature in black ink, enclosed within a roughly drawn, irregular oval border. The signature consists of a stylized 'F' followed by 'WLC'.

Frederik W. L. Christoffersen

Abstract

This thesis is in the workings... [\[Write something when finished with all thesis chapters\]](#)

Acknowledgements

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Generative AI tools were occasionally used as support for brainstorming, linguistic editing, code exploration, and research during the development of this project.

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Acronyms

HT	Haghia Triada Tablet
SAM	Segment Anything Model
mSAM	MobileSAMv2
Otsu	Otsu's method
GC+brush	GrabCut with brush seeding
mSAM+pts	MobileSAMv2 with automatic point prompts
Cefael	Collections de l'École française d'Athènes en ligne
GORILA	Godart and Olivier, Recueil des inscriptions en linéaire A
PDR	Project Definition Report
IPR	Intellectual Property Rights
GMM	Gaussian Mixture Model
IoU	Intersection over Union
Dice	Dice-Sørensen coefficient
TPE	Tree-structured Parzen Estimator

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CHAPTER 1

Introduction

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, lately consisting of only around 2,500 inscriptions (about 1,400 if only including those well-preserved enough to be readable) - significantly smaller than that of its younger counterpart that now spans more than 4,500 inscriptions (which are comparatively longer and better preserved).^{1,2} But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.^{3,4}

Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael) (see **EtudesCretoises21**), these are only available in print form. This

¹Salgarella2020.

²Salgarella2025.

³RILA-S1.

⁴CNRLinearACorpus2025.

motivated the development of the palæographical⁵ database *The Signs of Linear A* (SigLA), which seeks to "[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palæographic analyses within the epigraphic corpus" (**SalgarellaCastellanSigLA2020**). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known inscriptions.^{6,7}

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.⁸ According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation tool can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database.

1.1.2 A problem of small data

The small size of the Linear A corpus and the irregularities in quality of its documents is a huge problem for the application of deep learning approaches, which typically require large amounts of data to perform well. Consequently, this project will instead seek to make use of generalized

⁵The term "palæographical" refers to the study of ancient writing systems.

⁶**SigLAAboutPage**.

⁷**SigLASEarchDocuments**.

⁸**SalgarellaCastellanSigLA2020**.

machine learning and computer vision, in particular Meta’s state-of-the-art interactive segmentation tool, the Segment Anything Model (SAM) (**sam**), which is pre-trained on a massive and diverse dataset of over 1 billion masks. SAM’s generalization and interactivity (allowing user-placed markers that guide the model) might thus allow for semi-automatic annotation of Linear A inscriptions without the need for large amounts of training data, which is unfortunately not available in this context.⁹

⁹By the “interactivity” of SAM, what is meant is NOT any built-in user interface in the form of a GUI, but rather the possibility of prompt-based placement of rectangles of focus, and points of additions and deletions that guide the model’s prediction.

1.2 Prior work

This thesis places itself at the intersection of computer vision, digital humanities, and epigraphy (the study of inscriptions). Since it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. As a result, the other two fields will merely provide the context and motivation for the research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM, etc.), 2. the palæographical and digital humanities work that this project will seek to support (SigLA), and 3. the overall context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (**Salgarella2020**) and (**Salgarella2025**), its new addition (**RILA-S1**), and where it's accessed (**EtudesCretoises21**). And for the epigraphical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (**SalgarellaCastellanSigLA2020**), its about page (**SigLAAboutPage**), and its document repository (**SigLASEarchDocuments**). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (**DigitalImageProcessing**) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (**Otsu1979**), and of course the OpenCV

Python documentation ([opencv_imgproc_docs](#)) for short implementation reference. Then, for SAM, these are: its paper ([sam](#)), its lighter implementation MobileSAM ([mobile_sam](#)) - and the newest version of that one MobileSAMv2 ([mobile_sam_v2](#)). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art ([DeepLearningSegmentation](#)), and a lighter EU-funded historical document segmentation tool *dhSegment* ([dhsegment](#)) also using deep learning.

1.3 Problem statement

As previously stated, the aim of this project is to investigate whether parts of the process of creating annotated drawings for SigLA can be automated. More specifically, whether some kind of semi-automatic segmentation tool using interactive and generalized computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database. But what would be the relative segmentation quality of such a tool? How automatic would it be exactly? And by how much could it improve the efficiency of the workflow? For a decomposition of the problem statement into its three research questions - each accompanied by their respective operationalization - see the following subsection:

1.3.1 Research questions

1. **Segmentation performance:** How well does a SAM-based segmentation approach perform quantitatively in terms of segmentation accuracy, compared to simple classical image processing baselines when applied to Linear A document images? [\[Operationalization: SAM-based masks will be compared to classical image-processing baselines using overlap metrics \(e.g. IoU/Dice\) against the ground truth segmentations derived from SigLA's Krita files for a minimum of 25 documents.\]](#)

2. **Degree of automation:** To which extent can user interaction be reduced by a SAM-based semi-automatic segmentation workflow, while still maintaining high-quality outputs? [Operationalization: The number of user interactions (e.g. clicks) required to achieve a certain level of segmentation quality (e.g. IoU/Dice) will be measured against ground truth (SigLA's Krita files) for a minimum of 25 documents.]
3. **Workflow efficiency:** How much reduction in annotation time can be achieved using the proposed tool compared to fully manual digitization, under controlled experimental conditions? [Operationalization: The time it takes Ester Salgarella to annotate manually compared to being assisted by the proposed segmentation tool will be measured for a minimum of 5 documents.]

1.4 Overview of the thesis

The rest of this thesis is structured into five main chapters (Data, Methodology, Results, Discussion, Conclusion), all split up into sections concerning each of the three research questions.

CHAPTER 2

Data

2.1 Sources

The dataset used in this thesis was constructed from photographs of raw Linear A inscriptions, and their corresponding ground truth annotations. Cefael (**EtudesCretoises21**) was chosen as the source for raw document images, since it is the only online and widely publicly available source of the printed five volume corpus - Godart and Olivier, Recueil des inscriptions en linéaire A (GORILA). As all raw document images on the pages of this corpus are accompanied by annotated drawings, an argument could be made to make use of those as ground truth annotations. However, annotations provided by Ester Salgarella from SigLA (**SalgarellaCastellanSigLA2020**) were chosen instead, as they are more up to date, and are already in a binary mask format that is suitable for comparison with segmentation model outputs.

What follows in this chapter is a detailed description of this dataset, documenting its characteristics, its construction process, and its limitations and biases.

2.1.1 Characteristics and construction

The dataset was made to consists of 32 chosen image pairs of raw documents and their corresponding ground truth annotations. Only inscriptions with clay as writing support were included, as they are the most common and representative of the Linear A corpus, and are the only ones for which SigLA ground truth annotations were already available (IS THIS TRUE?). The raw document images from Cefael are screenshots of grayscale images within the scanned printed edition of the corpus, and are thus not of the highest quality, being the only widely publicly

available images of the Linear A inscriptions. For an example of a raw document image and its corresponding ground truth annotation, see Figure 2.1.

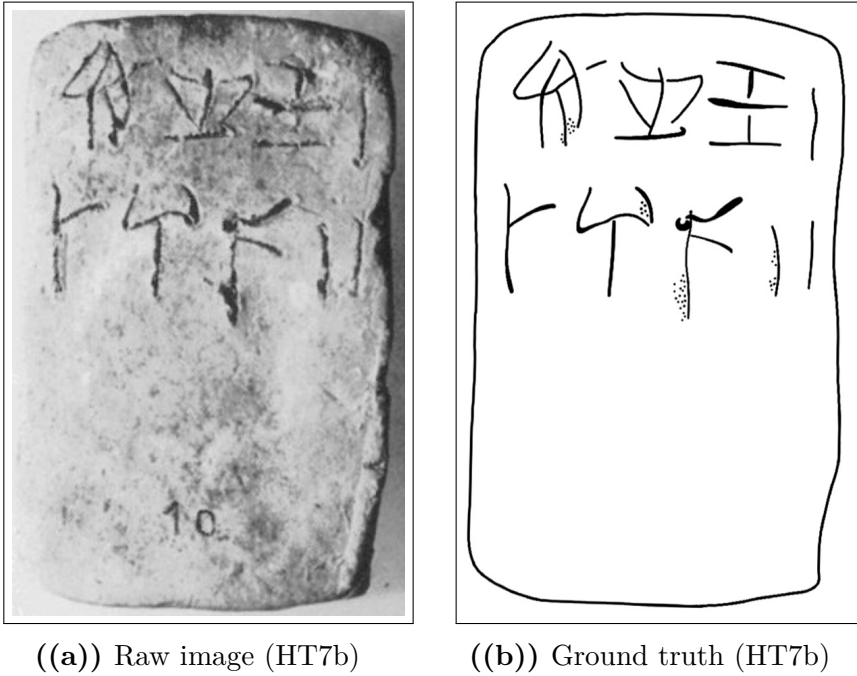


Figure 2.1: Example of a raw inscription image and its corresponding ground truth annotation.

2.1.2 Difficulty classification

The dataset was categorized into three difficulty levels (easy, medium, hard) based on the visual quality of the raw images and the clarity of the inscriptions. This classification was performed manually by inspecting each image and considering factors such as contrast, noise, and the presence of tablet breakages or general damage to the inscriptions. Concretely, the following criteria were used to classify each difficulty set (see Figure 2.2 for a summary of the criteria, and Figure 2.3 for examples of images with different difficulty levels):

X

- Easy: labeled as such for documents with clear engravings, good contrast, minimal noise, and constant lighting.
- Hard: labeled to those with poor contrast, significant noise, and substantial damage making it hard even for the naked eye to read the inscriptions.
- Medium: labeled to all those inbetween the qualities of easy and hard.

Figure 2.2: Criteria for classifying the dataset into easy, medium, and hard difficulty levels.

2.1.3 Image registration and preprocessing

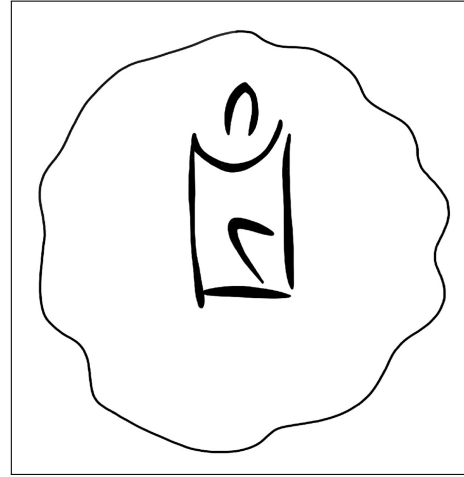
To obtain any kind of meaningful comparison between model outputs and ground truth, the raw images and their ground truth annotations first had to be aligned and registered to each other. This was done manually in Krita by downscaling the ground truth images to the individual scales of their raw counterparts, and aligning by translation and rotation, using the raw images as background reference.

2.2 Limitations and biases

Even after image registration, ground truth images still did not perfectly align with the raw images, affecting the obtained segmentation quality. This misalignment is a source of noise in the evaluation.



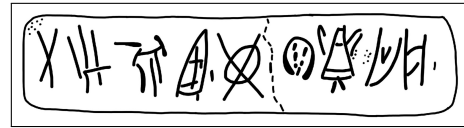
((a)) Easy KHWc2104 (raw)



((b)) Easy KHWc2104 (gt)



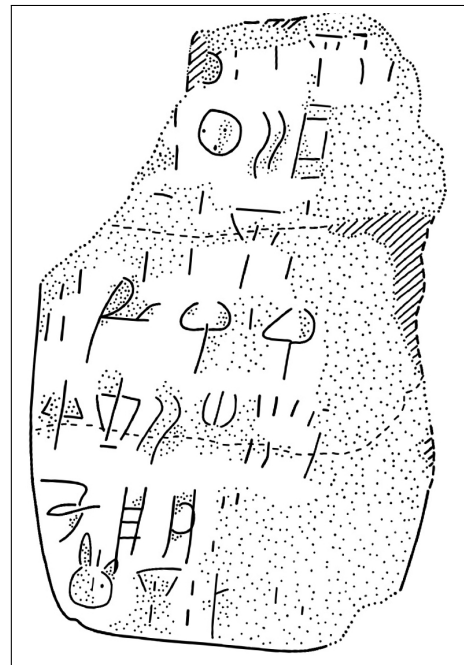
((c)) Medium MA1a (raw)



((d)) Medium MA1a (gt)



((e)) Hard HT3 (raw)



((f)) Hard HT3 (ground truth)

Figure 2.3: Examples of images with different levels of segmentation difficulty: easy, medium, and hard. For each case, the raw image (left) and the corresponding ground truth annotation (right) are shown.

CHAPTER 3

Methodology

3.1 Common methodology

3.1.1 Image processing theory

[SHOULD THIS SECTION EXIST? IF SO, I SHOULD FINISH IT]

The basics of spatial filtering can be described by the following equation:

$$g(x, y) = T[f(x, y)]$$

, where $f(x, y)$ (also denoted as $I(x, y)$) is the input image, $g(x, y)$ is the output image.¹

3.1.1.1 Binary image thresholding

$$B(x, y) = \begin{cases} 1 & \text{if } I(x, y) > T \\ 0 & \text{otherwise} \end{cases}$$

2

3.1.1.2 Convolutional kernels

3

$$(w \star I)(x, y) = \sum_{i=-a}^a \sum_{j=-b}^b w(i, j) I(x - i, y - j)$$

¹DigitalImageProcessing.

²DigitalImageProcessing.

³DigitalImageProcessing.

where w is the convolution kernel (window) of size $m \times n$, and $a = (m-1)/2$ and $b = (n-1)/2$ are the half-width and half-height (assuming odd dimensions), respectively.⁴

3.1.2 Development environment and tools

[TODO: EXTEND THIS SECTION] Python 3.10 was used for all code development (because it has such a big library of image processing functions and machine learning model implementations), with the following libraries:

- OpenCV for image processing and computer vision tasks.
- NumPy for numerical operations and array manipulations.
- Matplotlib for data visualization.
- *Add more!*

Krita was used for manual image registration, making sure the raw image ground truth pairs were properly aligned.

3.1.3 General evaluation metrics

3.2 Segmentation quality

To answer the question of RQ1: *How does the one-shot segmentation quality of SAM compare to that of classic segmentation methods?*, the one-shot segmentation quality of a computationally efficient implementation of SAM (MobileSAMv2) was evaluated against a set of baseline models, using a suitable segmentation metric (i.e. Dice score) computed from the pairs of predicted and ground truth masks. All models were manually tuned to their best performance on the dataset, and the best performing version of SAM was compared against the best performing baseline model.

⁴`DigitalImageProcessing`.

The following subsections will therefore document the configuration of the baseline and state-of-the-art models, as well as the evaluation protocol and performance metrics used to answer RQ1.

3.2.1 Baseline model configuration

The baseline models against which the best SAM implementation was evaluated (from simplest to most advanced) were:

- Global thresholding (Otsu’s method)
- Local adaptive thresholding (Gaussian) with bilateral pre-filtering
- Edge-based segmentation (Canny edge detection with region filling)
- Energy-based segmentation (GrabCut with brushseeds)

Together, they cover a wide range of classic segmentation techniques that might be used for binary foreground extraction. What follows is a brief description of each of these methods, including their characteristics, how they work, and most importantly their implementation details.

3.2.1.1 Global thresholding (Otsu’s method)

Otsu’s method is a global thresholding method that works on gray-level histograms, which was originally proposed by **Otsu1979**. As the simplest of the baselines, it represents the minimum acceptable segmentation quality, since it only takes the global gray level distribution into account, with no assumptions about any spatial relationships between pixels. It is therefore expected to perform poorly on images with erosion, uneven lighting, and differences in surface texture⁵, which are the exact characteristics of the images to be segmented.

Here is how it works: First, let pixels of the raw image be divided into L shades of gray $[1, 2, \dots, L]$. Then, let each pixel be assigned one of two classes C_0 or C_1 , guided by a threshold T , where C_0 contains the shades

⁵**DigitalImageProcessing**.

$[1, 2, \dots, T]$, and C_1 contains the rest $[T + 1, T + 2, \dots, L]$. Otsu's method then works by performing an exhaustive search over all L gray levels to find an optimal threshold T^* that results in a maximum between-class variance σ_B^2 :

$$T^* = \arg \max_{1 \leq T < L} \sigma_B^2(T)$$

The between-class variance σ_B^2 is then defined by a function of the respective total probabilities of the two classes (ω_0 and ω_1) and the mean gray level value per class (μ_0 and μ_1):

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T)[\mu_1(T) - \mu_0(T)]^2$$

3.2.1.2 Local adaptive thresholding (Gaussian) with bilateral pre-filtering

Local adaptive thresholding works by computing a threshold for each pixel based on its local neighborhood. This is an advantage in cases where the image has varying light conditions (which is the case for the images to be segmented), as the threshold adapts to local lighting.

Mathematically, the local threshold T is a function of a given pixel position (x, y) computed by a weighted sum of the neighborhood. The generated binary mask B is then defined as:⁶

$$B(x, y) = \begin{cases} 1 & \text{if } I(x, y) > T(x, y) \\ 0 & \text{otherwise} \end{cases}$$

To compute $T(x, y)$, there are a few weighting functions to choose from. For this baseline, the Gaussian kernel was used, as it can limit the influence of distant noisy pixels by assigning weights based on distance from the kernel center. With a Gaussian kernel G_σ , the threshold $T(x, y)$ is computed as a Gaussian-weighted sum of local intensities subtracted by a constant C . Using convolution ($\star$), this can be expressed as:⁷

$$T(x, y) = (G_\sigma \star I)(x, y) - C$$

⁶`DigitalImageProcessing`.

⁷`opencv_adaptive_threshold`.

In general, the weight at an offset (i, j) from the kernel center can be written as follows (where α is a normalization constant ensuring that the kernel sums to 1, and σ controls the spatial spread of the kernel):⁸

$$G_\sigma(i, j) = \alpha \cdot \exp\left(-\frac{i^2 + j^2}{2\sigma^2}\right)$$

As a pre-filtering step, a bilateral filter (**TomasiManduchi1998**) was applied to the input image before performing the Gaussian thresholding. In this way, more noise could be reduced while preserving inscriptions, as this filter is known to be effective at reducing noise while preserving edges. It takes three hyperparameters: the kernel window diameter d , the spatial standard deviation σ_{space} , and the intensity standard deviation σ_{color} . Discretely, its output intensity $I_{\text{bf}}(p)$ at a given pixel position p (where q is the position of one of its neighbors within the kernel window) can be defined as follows:⁹

$$I_{\text{bf}}(p) = \frac{\sum_q I(q) W_{\text{space}}(p, q) W_{\text{color}}(p, q)}{\sum_q W_{\text{space}}(p, q) W_{\text{color}}(p, q)}$$

Here, the spatial weight based on the distance between pixels p and q , and the color range weight based on the intensity difference between p and q are respectively given by:

$$W_{\text{space}}(p, q) = \exp\left(-\frac{\|p - q\|^2}{2\sigma_{\text{space}}^2}\right)$$

$$W_{\text{color}}(p, q) = \exp\left(-\frac{\|I(p) - I(q)\|^2}{2\sigma_{\text{color}}^2}\right)$$

3.2.1.3 Edge-based segmentation (Canny edge detection with region filling)

Canny edge detection is a method that by itself only segments the edges of an image, excluding the enclosed regions within those edges. Here is

⁸**DigitalImageProcessing.**

⁹**Pan2025InnovativeBilateral.**

how it works: First, the input image is smoothed using a Gaussian filter to reduce noise (using the Gaussian kernel as defined in the previous section):

$$I_s(x, y) = (G_\sigma \star I)(x, y)$$

Then, for the image gradient of each pixel, the magnitude $M_s(x, y)$ and the orientation $\theta_s(x, y)$ are computed:¹⁰

$$M_s(x, y) = \|\nabla I_s(x, y)\| = \sqrt{\left(\frac{\partial I_s}{\partial x}\right)^2 + \left(\frac{\partial I_s}{\partial y}\right)^2}$$

$$\theta_s(x, y) = \arctan\left(\frac{\partial I_s / \partial y}{\partial I_s / \partial x}\right)$$

Next, *non-maxima suppression*¹¹ is applied to retain only local maxima of $M_s(x, y)$ along the gradient direction $\theta_s(x, y)$, resulting in thinner edges. This is at last followed by *hysteresis thresholding*¹², classifying pixels as strong, weak, or non-edges with a low threshold T_l and a high threshold T_h , only preserving weak edges if connected to strong edges.

In an attempt to fill the regions enclosed by the detected edges, morphological closing (dilation followed by erosion) was applied to the edge mask.

3.2.1.4 Energy-based segmentation (GrabCut with brushseeds)

GrabCut (**Rother2004GrabCut**) is a classical interactive segmentation method, which makes for a good comparison to a deep learning interactive model like SAM. It is good at producing spatially coherent masks with little large-grained noise, which is ideal for the images to be segmented.

The algorithm is interactively initialized by the user, who traditionally provide a bounding box around the object of interest. Pixels outside this box are marked as definite background, while pixels inside it are

¹⁰**DigitalImageProcessing.**

¹¹**DigitalImageProcessing.**

¹²**DigitalImageProcessing.**

marked as initially unknown. To further guide the segmentation, the user also has the option to provide an additional number of pixels as definite foreground ($B_n = 1$), and definite background ($B_n = 0$).

As an energy-based segmentation method, optimal segmentation is achieved by minimizing a cost (energy) function E that combines a data term D and a smoothness term S , given an input image I , a binary segmentation mask B , and a set of parameters θ :¹³

$$E(B, I, \theta) = D(B, I, \theta) + S(B, I)$$

The set of parameters θ then define two *Gaussian Mixture Models* (GMMs) (probabilistic models each consisting of a weighted sum of several Gaussian distributions) that over iterations will be tuned to model the foreground and background pixel distributions. These are used to compute the data term D whose energy contribution decreases the better B matches the GMMs of foreground and background. The smoothness term S then penalizes neighboring pixels that have different labels, encouraging spatial coherence in the segmentation.

At last, the optimal segmentation B^* is found iteratively, by alternating between optimizing the GMM parameters θ to model B , and minimizing the energy function with respect to B :

$$B^* = \arg \min_B E(B, I, \theta)$$

The automatic brush seeds (see Figure 3.1 for a visualization) given to GrabCut were derived from the thresholded mask of the Gaussian adaptive thresholding baseline (see Section 3.2.1.2), as it seems to be the best at extracting foreground from the Linear A tablets, albeit with a degree of noise. The erosion of the thresholded mask was given as sure foreground, while the erosion of the inverse of the thresholded mask was given as sure background, and the rest was given as probable background. In that way, the model was provided with a good amount of guidance, while still leaving room for the model to get rid of noise and fill in gaps within the engravings of the inscriptions.

¹³This is a simplified version of the energy function that is used by GrabCut.

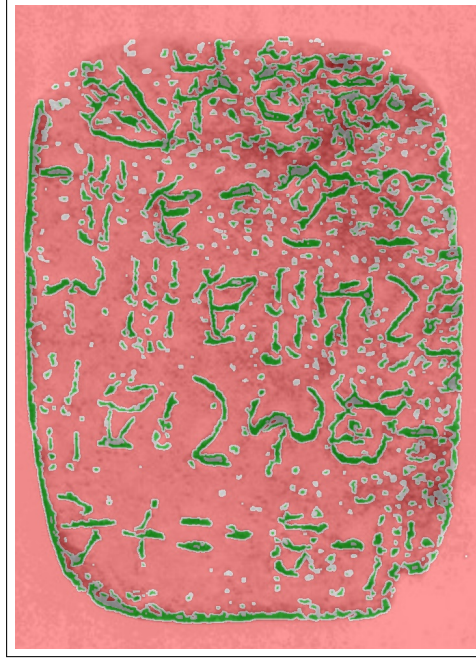


Figure 3.1: A visualization of seed distribution for GrabCut with brush seeding (GC+brush) on HT118. Green overlay represents sure foreground pixels, while red overlay represents sure background pixels, and absence of overlay represents probable background pixels.

3.2.2 State-of-the-art model configuration

3.2.2.1 Model architecture

Raw image $\rightarrow$ Automatic prompts $\rightarrow$ MobileSAMv2 $\rightarrow$ Binary mask

3.2.2.2 Segmentation (MobileSAMv2)

For the sake of computational efficiency, a mobile implementation of SAM - MobileSAMv2 (mSAM) (**mobile_sam_v2**) was used instead of the full SAM model. This is a lightweight version of the original SAM model, which is designed to run efficiently on mobile devices while still maintaining good segmentation performance. It is trained on a smaller sample of the same dataset as the original SAM, and thus has roughly the

same one-shot segmentation capabilities, although with a slightly lower performance than the original.

Here is how it works: ViT encoder, prompt encoder, mask decoder, etc.

[STILL YET TO DOCUMENT HOW IT WORKS]

3.2.2.3 Seed extraction

Even though the mSAM implementation already makes use of an automatic box prompt, it does so from a YOLO model trained on the same dataset as SAM. Those box prompts are generalized for and work best on natural images. Hence, they may not perform well on gray-scale images of clay tablets with Linear A inscriptions, as those were not part of the training data. Thus, mSAM needed to be adapted through the use of prompts. The model takes two types of prompts: 1. a single box prompt, and/or 2. point prompts labeled as either foreground or background. For the one-shot segmentation capability of the model, these prompts had to be automatically generated, rather than interactively placed by a user. However, as it is particularly difficult to automatically create high quality box prompts without a trained object detection model, only point prompts were used.. Thus, two versions of MobileSAMv2 were made for comparison:

- **mSAM**: MobileSAMv2 (with the original automatic box prompts from the YOLO model),
- and **mSAM+pts**: MobileSAMv2 with automatic point prompts labeled as either foreground or background.

Just like in the case of GC+brush, the automatic point prompts for mSAM+pts (see Figure 3.3 for visualization) were derived from thresholded masks of the Gaussian adaptive thresholding baseline (see Section 3.2.1.2), This was done by randomly sampling a number of foreground and background pixels from the mask as seeds for mSAM+pts. More specifically, the foreground seeds were sampled from the foreground of one Gaussian thresholded mask, while the background seeds were sampled from the erosion of the inverse of a second Gaussian thresholded

mask. The first erosion of the background was This gave 9 hyperparameters (see Figure 3.2) that were tuned for optimal performance of mSAM+pts:

3.2.3 Hyperparameter optimization

To ensure a fair comparison between the models, all models with hyperparameters were tuned to their best performance on the dataset using Optuna, a hyperparameter optimization framework. The optimization itself was performed using the Tree-structured Parzen Estimator (TPE) algorithm (**TPEHyperparameterOptimization2013**)¹⁴, which is a Bayesian optimization method that models the relationship between hyperparameters and performance to efficiently explore the hyperparameter space and find the optimal ones. Hence, it is more sophisticated than a random search, which just samples different hyperparameter combinations without considering the results of past evaluations.

It works by building two probabilistic models: one for the hyperparameter configurations that have resulted in good performance, and another for those that have resulted in poor performance. Then, it uses these models to select the next hyperparameter configuration to evaluate, by maximizing the expected improvement over the current best performance. This way, it can focus the search on promising regions of the space of possible hyperparameters without having to rely on chance or exhaustive search.

3.2.4 Evaluation protocol and performance metrics

For an evaluation of one-shot segmentation quality, relevant metrics and statistical tests needed to be defined to determine whether one model performed significantly better than another. More specifically, it was of particular interest to compare the best performing baseline model against the best performing version of MobileSAMv2.

¹⁴**TPEHyperparameterOptimization2011.**

X

1. C : the constant subtracted from the Gaussian-weighted sum to compute the local threshold (see Section 3.2.1.2),
2. σ : the spatial standard deviation of the Gaussian kernel used for adaptive thresholding (see Section 3.2.1.2),
3. $d_{bilateral}$: the diameter of the kernel window for bilateral pre-filtering (see Section 3.2.1.2),
4. σ : the spatial and intensity standard deviation for bilateral pre-filtering (see Section 3.2.1.2),
5. n_{fgd_points} : the number of foreground point seeds to be sampled from the foreground of the first Gaussian thresholded mask,
6. $n_{boundary_points}$: the number of background point seeds to be sampled from the boundary region between foreground and sure background,
7. $n_{sure_bgd_points}$: the number of background point seeds to be sampled from the sure background region,
8. $d_{gap_erosion}$: the kernel size for erosion used to create a gap between the foreground and background regions.
9. $d_{boundary_erosion}$: the kernel size for erosion used to determine the size of the boundary region.

Figure 3.2: Hyperparameters for MobileSAMv2 with automatic point prompts (mSAM+pts)

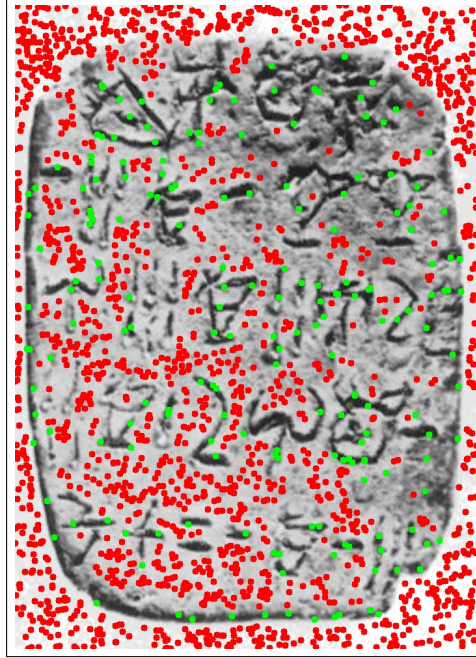


Figure 3.3: A visualization of seed distribution for MobileSAMv2 with automatic point prompts (mSAM+pts), overlaid on HT118. Green points represent foreground seeds, while red points represent background seeds.

The following subsections will therefore document the choice of metrics and statistical tests for the evaluation, used to answer RQ1: *How does the one-shot segmentation quality of SAM compare to that of classic segmentation methods?*

3.2.4.1 Choice of metrics

When it comes to evaluation of segmentation models, the two most commonly used metrics are: 1. the Dice-Sørensen coefficient (Dice) and 2. the Intersection over Union (IoU). The Dice score is most commonly used in cases where there are heavy class imbalances (e.g. lots of background compared to regions segmented), and when foreground objects are small and thin. It is naturally more forgiving than the related seg-

mentation metric IoU, which instead tends to be rather strict, and is specifically used to evaluate segmentation results where small displacements are critical for results. Therefore, segmentation performance was reported using Dice. Mathematically, it is defined as twice the size of the intersection of the predicted segmentation mask $\hat{Y}$ and the ground truth mask Y , divided by their summed size:¹⁵

$$\text{Dice}(\hat{Y}, Y) = \frac{2 \cdot |\hat{Y} \cap Y|}{|\hat{Y}| + |Y|}$$

Thus, the raw metrics ended up consisting of the Dice score for each image, and each model.

3.2.4.2 Statistical tests

To determine whether the observed differences in scores between models were statistically significant, paired t-tests were conducted. The paired t-test has two main assumptions: 1. independence of observations, and 2. normality of residuals. The first assumption is already satisfied a priori, since the segmentation performance of one image does not directly influence the performance on another. However, the second assumption depends on results. Thus, the residuals of the paired differences in Dice scores between the two models in question needed to be checked for normality using a QQ-plot and the Shapiro-Wilk test. In case the normality assumption was violated, the non-parametric domain-specific Wilcoxon signed-rank test would be used instead.¹⁶

3.3 Interactive semi-automation

Here, the interactivity and quality of human-guided SAM (and GrabCut for comparison?) will be evaluated against the one-shot results from Section 3.2. [Add some context of RQ2 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

¹⁵**Metrics2025.**

¹⁶**Rainio2024.**

3.3.1 Interactive tool architecture

3.3.2 Experimental design for automation evaluation

3.3.3 Definition of automation metrics

3.4 Workflow efficiency

[Add some context of RQ3 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

3.4.1 Export to Krita

[It is indeed possible via the Krita Python API to export the predicted segmentation masks as Krita layers, which can then be further edited by hand by a user. The question is only if I want to do so via the API (requires Krita installed), or via a Python export script. Krita's file format is open, so both are possible.]

3.4.2 Definition of workflow efficiency metrics

3.4.3 Experimental design of the user study

3.4.4 Analysis of results

CHAPTER 4

Results and Discussion

4.1 Segmentation ability (one-shot)

After running the evaluation of the models on the dataset as illustrated in Figure 4.1, it could already be seen that no implementation of MobileSAMv2 (mSAM) was in the lead, while GC+brush seemed to outperform all. See the appendix for the full results table, and the next section for a more detailed analysis of the results.

4.1.1 Statistical significance

4.1.1.1 Normality of residuals

As the QQplot in Figure 4.3 shows, the residuals of the paired differences in Dice scores between GC+brush and mSAM+pts appear to be approximately normally distributed, which supports the validity of the paired t-test results. Furthermore, the Shapiro-Wilk test for normality on these residuals yields a p-value of $1.55 \cdot 10^{-1} > \alpha = 0.05$, indicating that we fail to reject the null hypothesis of normality.

4.1.1.2 Paired t-test

To determine whether the classic baseline model GC+brush significantly outperformed mSAM+pts, a paired t-test is conducted on the Dice scores obtained for each image across these two models. The null hypothesis (H_0) states that there is no difference in mean Dice scores between the two models, while the alternative hypothesis (H_1) states that GC+brush has a higher mean Dice score than mSAM+pts. Performing the paired

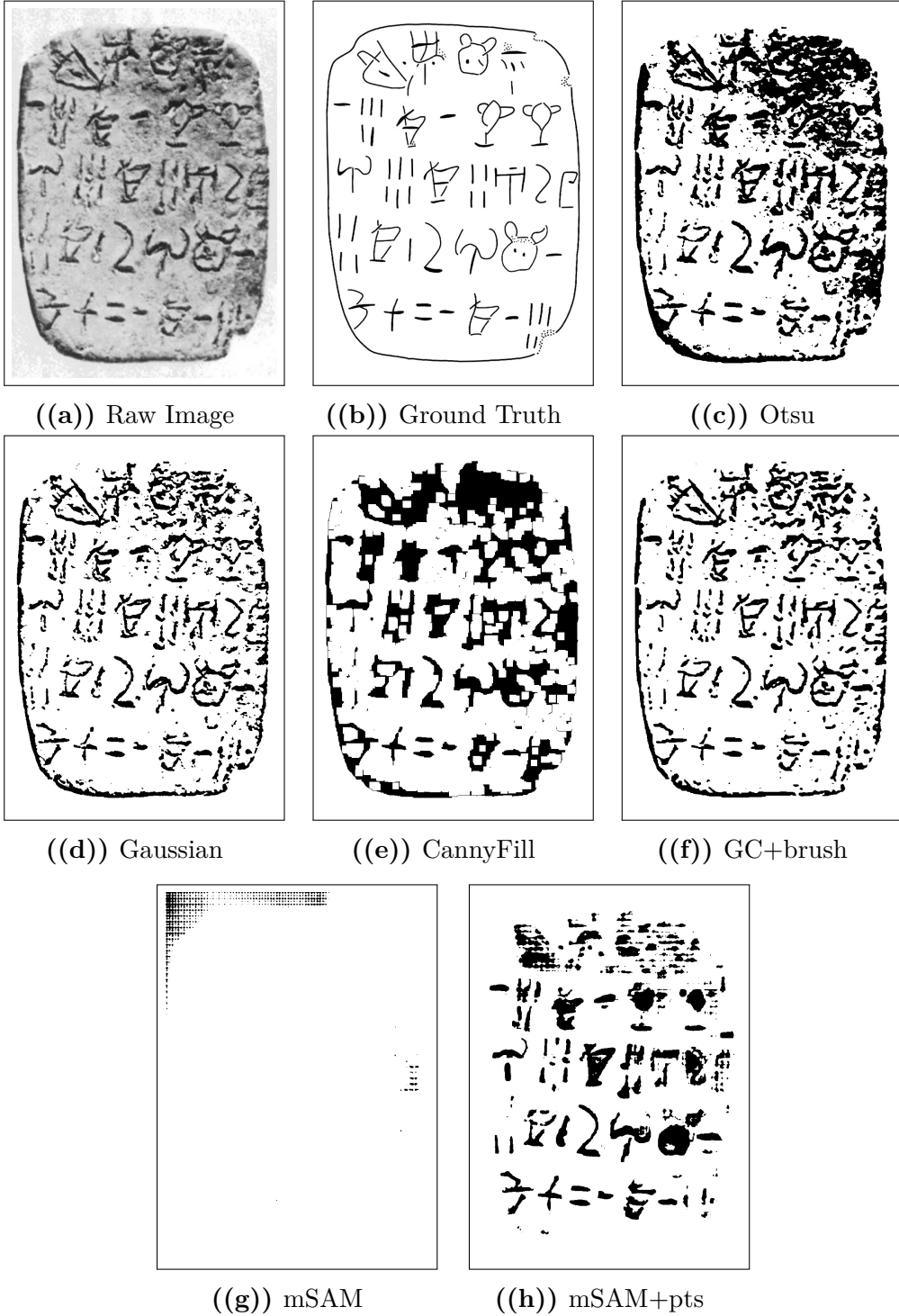


Figure 4.1: Comparison of segmentation outputs from different models against the raw image and ground truth.

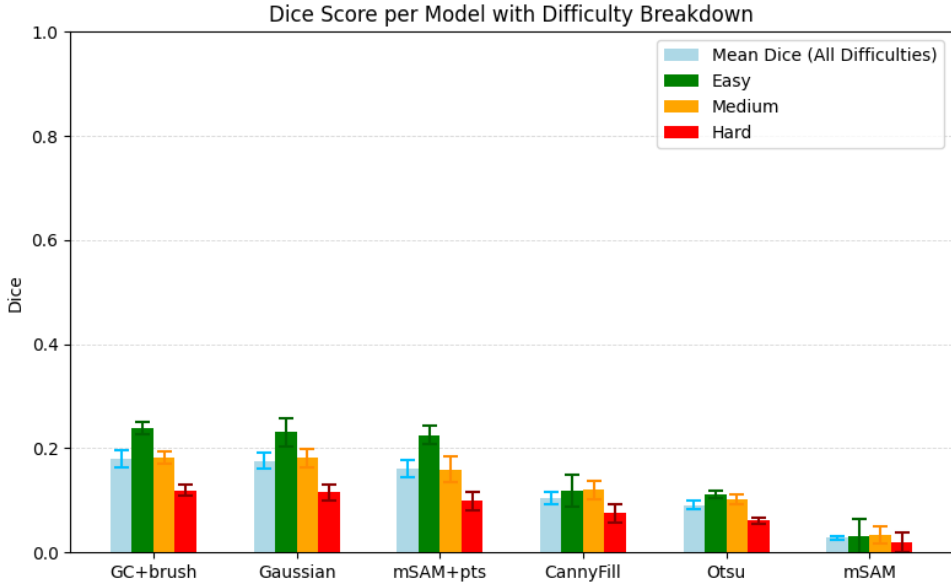


Figure 4.2: Dice Score performance per model. Split by difficulty. With error bars representing the standard error of the mean. Ordered descendingly from the left by mean performance.

t-test yielded the following results:

$$\delta = \mu_1 - \mu_2 \approx 0.18 - 0.161 = 0.02$$

$$t \approx 3.642$$

$$P(T > t) \approx 1.048 \cdot 10^{-3}$$

Thus, as $1.048 \cdot 10^{-3} < \alpha = 0.05$, the null hypothesis can be rejected, and it can be concluded that GC+brush significantly outperforms mSAM+pts in terms of Dice score. Hence, it can be said that mSAM underperforms in comparison to classical segmentation methods for one-shot segmentation of Linear A tablets, even when automatically prompted.

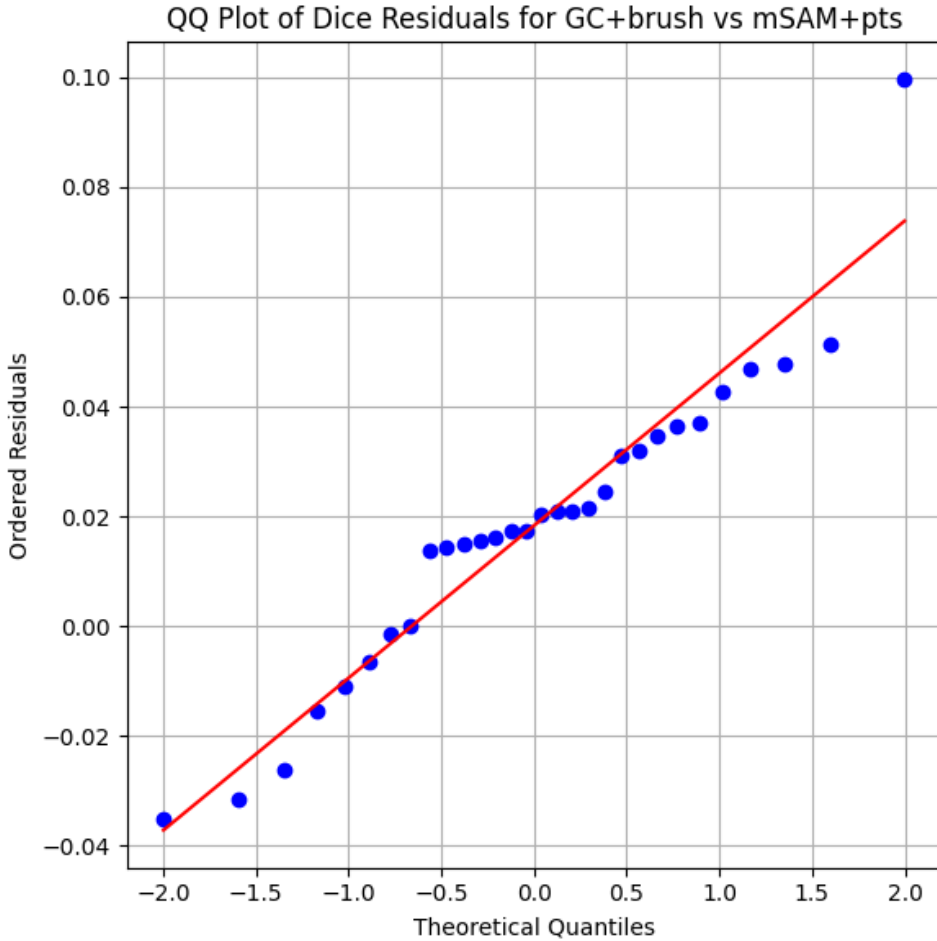


Figure 4.3: QQplot of the residuals of the paired t-test comparing GC+brush and mSAM+pts. The residuals appear to be approximately normally distributed, which supports the validity of the paired t-test results. Furthermore, the Shapiro-Wilk test for normality on these residuals yields a p-value of $1.55 \cdot 10^{-1} > \alpha = 0.05$, indicating that we fail to reject the null hypothesis of normality.

4.2 Automation and interactivity

4.3 Workflow impact

CHAPTER 5

Conclusion

5.1 Summary of findings per research question

5.1.1 RQ1: Segmentation performance

5.1.2 RQ2: Degree of automation

5.1.3 RQ3: Workflow efficiency

5.2 Overall contribution

5.3 Future Work

5.3.1 Methodological extensions

5.3.2 Dataset expansion and diversification

5.3.3 Further automation opportunities

APPENDIX A

An Appendix

A.1 Raw segmentation performance

Table A.1: Raw Dice score data on easy images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT10a	0.179	0.254	0.254	0.131	0.057	0.24
HT118	0.125	0.156	0.157	0.109	0.002	0.132
HT12	0.148	0.267	0.261	0.192	0.048	0.215
HT17	0.158	0.23	0.225	0.071	0.03	0.265
HT22	0.024	0.191	0.191	0.136	0.057	0.17
HT26a	0.14	0.261	0.264	0.116	0.001	0.225
HT7a	0.115	0.127	0.133	0.116	0.02	0.159
HT7b	0.168	0.17	0.168	0.074	0.026	0.133
KHWc2104	0.005	0.482	0.439	0.069	0.027	0.497
PH2	0.12	0.24	0.225	0.096	0.046	0.224

Table A.2: Raw Dice score data on medium images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT154B	0	0.136	0.152	0.073	0.032	0.121
HT16	0.162	0.173	0.167	0.081	0.003	0.179
HT2	0.071	0.114	0.117	0.094	0.021	0.093
HT4	0.11	0.237	0.24	0.157	0.061	0.216
HT40	0.122	0.154	0.161	0.088	0.029	0.119
HT6b	0.122	0.182	0.178	0.075	0.004	0.168
HT85b	0.124	0.113	0.104	0.06	0.035	0.095
HT8b	0.237	0.281	0.283	0.11	0.052	0.181
HT90	0.018	0.189	0.173	0.14	0.012	0.174
MA1a	0.238	0.246	0.243	0.144	0.09	0.247

Table A.3: Raw Dice score data on hard images. Label column represents the name of the document segmented, while all other columns represent Dice scores obtained for those documents for a given model.

Label	CannyFill	GC+brush	Gaussian	Otsu	mSAM	mSAM+pts
HT1	0.128	0.124	0.114	0.043	0.049	0.082
HT3	0.079	0.087	0.085	0.058	0.046	0.066
HT30	0.098	0.101	0.1	0.054	0.006	0.101
HT33	0.035	0.05	0.049	0.016	0	0.033
HT52a	0.002	0.103	0.098	0.065	0.006	0.071
HT85a	0.115	0.099	0.1	0.057	0.006	0.125
HT8a	0.177	0.205	0.203	0.061	0.041	0.158
HT94b	0.064	0.081	0.078	0.038	0.017	0.05
HT9a	0.021	0.223	0.208	0.147	0.02	0.234
HT9b	0.032	0.121	0.116	0.071	0	0.074

Table A.4: Summary Statistics per Model of Dice Score on all images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.105	0.066	0.012	30
GC+brush	0.18	0.086	0.016	30
Gaussian	0.176	0.08	0.015	30
Otsu	0.091	0.041	0.007	30
mSAM	0.028	0.023	0.004	30
mSAM+pts	0.161	0.09	0.016	30

Table A.5: Summary Statistics per Model of Dice Score on easy images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.118	0.058	0.018	10
GC+brush	0.238	0.098	0.031	10
Gaussian	0.232	0.086	0.027	10
Otsu	0.111	0.038	0.012	10
mSAM	0.031	0.02	0.006	10
mSAM+pts	0.226	0.106	0.033	10

Table A.6: Summary Statistics per Model of Dice Score on medium images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.12	0.079	0.025	10
GC+brush	0.182	0.057	0.018	10
Gaussian	0.182	0.057	0.018	10
Otsu	0.102	0.034	0.011	10
mSAM	0.034	0.028	0.009	10
mSAM+pts	0.159	0.051	0.016	10

Table A.7: Summary Statistics per Model of Dice Score on hard images.

Model	Mean Dice	Standard deviation	Standard error	n
CannyFill	0.075	0.055	0.017	10
GC+brush	0.119	0.054	0.017	10
Gaussian	0.115	0.051	0.016	10
Otsu	0.061	0.034	0.011	10
mSAM	0.019	0.019	0.006	10
mSAM+pts	0.099	0.06	0.019	10

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